

Foundation Models for Satellite Earth Observation: A Comprehensive Survey of Architectures, Pre-training Strategies, and Deployment Challenges

Abstract

The advent of Foundation Models (FMs) has precipitated a paradigm shift in Satellite Earth Observation (EO), transitioning the field from task-specific, supervised learning toward generalizable, self-supervised representation learning. This survey provides a comprehensive synthesis of the current landscape of Geospatial Foundation Models (GFMs), analyzing over 300 recent academic contributions. We systematically categorize methodological advancements across three primary dimensions: architectural innovations adapting Vision Transformers, State Space Models, and diffusion processes to multi-spectral and multi-temporal data; pre-training strategies encompassing Masked Image Modeling, contrastive learning, and cross-sensor alignment; and deployment frameworks addressing onboard inference, parameter-efficient fine-tuning, and operational robustness. Our analysis reveals that while scaling laws generally hold for EO data, domain-specific adaptations such as wavelength-conditioned hypernetworks and spectral-aware masking are critical for handling the heterogeneity of satellite sensors. Furthermore, we identify significant gaps in out-of-distribution generalization, physical consistency, and the integration of vector data, outlining a roadmap for future research in agentic geospatial intelligence and physics-informed foundation modeling.

1. Introduction

Earth Observation (EO) has historically been constrained by the “curse of dimensionality” inherent in multi-spectral, hyper-spectral, and temporal data, coupled with a severe scarcity of high-quality labeled annotations. Traditional deep learning approaches, reliant on supervised training from scratch or transfer learning from natural image domains (e.g., ImageNet), often struggle to capture the complex spectral signatures and spatiotemporal dynamics unique to remote sensing [1, 2]. The emergence of Foundation Models—large-scale neural networks pre-trained on vast corpora of unlabeled data via self-supervised objectives—offers a transformative solution. By learning universal representations from petabyte-scale archives such as MajorTOM, SSL4EO, and Akupara, these models enable downstream task adaptation with minimal labeled data, effectively democratizing access to advanced geospatial analytics [3, 4, 5].

The evolution of GFMs has progressed rapidly from early adaptations of computer vision architectures to sophisticated, sensor-agnostic systems capable of processing heterogeneous data streams. Initial efforts focused on applying Masked Autoencoders (MAE) and contrastive learning frameworks like MoCo and DINO to single-sensor optical imagery [6, 7]. However, the unique characteristics of EO data—specifically the availability of multi-sensor views (e.g., Sentinel-1 SAR and Sentinel-2 Optical) of the same location, fine-grained temporal sampling, and explicit geospatial metadata—have driven the development of novel pre-training paradigms. Methods such as CROMA, DOFA, and Panopticon leverage these intrinsic properties to learn robust, invariant representations that generalize across sensors, seasons, and geographic regions [8, 9, 10].

Despite these advances, significant challenges remain. Recent benchmarks indicate that many GFMs fail to demonstrate superior robustness to distribution shifts compared to generic models, raising questions about the efficacy of current pre-training objectives [11, 12]. Furthermore, the computational demands of billion-parameter models necessitate innovations in model compression,

quantization, and onboard deployment for real-time applications [13, 14]. This survey synthesizes the state-of-the-art in GFMs, providing a detailed taxonomy of methodologies, evaluating benchmark performance, and identifying critical open problems in the pursuit of Artificial General Intelligence for the Earth.

2. Method

The methodological landscape of Foundation Models for Earth Observation is diverse, spanning architectural innovations, novel self-supervised objectives, and specialized adaptation techniques. We organize this section into coherent subsystems reflecting the primary research directions: Architectural Backbones, Pre-training Paradigms, Multi-Modal and Cross-Sensor Fusion, Temporal Modeling, Parameter-Efficient Adaptation, and Generative World Models.

2.1 Architectural Backbones for Geospatial Data

The backbone architecture serves as the computational engine for GFMs, determining the model’s capacity to capture spatial, spectral, and temporal dependencies. While Vision Transformers (ViTs) have become the de facto standard, recent works explore hybrid CNN-Transformer designs and emerging State Space Models (SSMs) to address specific EO constraints.

2.1.1 Vision Transformers and Hierarchical Variants Vision Transformers, with their global receptive fields, are particularly well-suited for capturing long-range dependencies in satellite imagery, such as large-scale land cover patterns and atmospheric phenomena. Standard ViT architectures have been scaled to billions of parameters, demonstrating consistent performance gains in rotated object detection and semantic segmentation tasks [15]. However, the quadratic complexity of self-attention limits their application to ultra-high-resolution inputs. To mitigate this, hierarchical architectures like Swin Transformers and their variants (e.g., SatSwinMAE) introduce shifted-window attention mechanisms, reducing complexity to linear while preserving multi-scale feature fidelity [16, 17].

Specialized adaptations further enhance ViT performance in EO contexts. For instance, the *Any-Optical-Model* (AOM) introduces a Spectrum-independent Tokenizer (SiTok) and Multi-scale Adaptive Patch Embedding (MAPE), enabling a single backbone to process arbitrary band configurations and spatial resolutions without retraining [18]. Similarly, *Scale-ALiBi* incorporates Ground Sample Distance (GSD) directly into the attention bias terms, allowing transformers to natively align tokens from images of varying resolutions [19]. These innovations underscore the necessity of embedding physical sensor parameters into the architectural inductive biases of GFMs.

2.1.2 State Space Models and Linear Complexity Architectures Addressing the computational bottlenecks of transformers, recent research has integrated State Space Models (SSMs), specifically the Mamba architecture, into EO backbones. *SatMamba* replaces standard ViT blocks with multi-way Mamba blocks that support four scanning directions (row/column major, forward/backward), achieving linear computational scaling while maintaining competitive representation quality for semantic segmentation [20]. Building on this, *RoMA* employs an autoregressive pre-training framework tailored for Mamba’s sequential scanning nature, incorporating adaptive rotation encoding to handle the orientation variance typical in overhead imagery [21]. Furthermore, *PathMamba* utilizes a hybrid Mamba-Transformer backbone, strategically placing Mamba blocks in early and late stages to prioritize topological continuity in road segmentation tasks, outperforming pure transformer baselines in preserving path connectivity [22]. These developments suggest that

SSMs offer a promising trajectory for scaling GFMs to continental-level analysis with constrained computational resources.

2.1.3 Hybrid CNN-Transformer and Lightweight Designs While transformers excel at global context, Convolutional Neural Networks (CNNs) retain advantages in capturing local texture and high-frequency details with lower data requirements. Hybrid architectures seek to combine these strengths. *Temporal Hybrid Swin-CNN* processes 4D spatiotemporal tensors using parallel high-frequency CNN and low-frequency Swin Transformer branches, effectively capturing both textural dynamics and global temporal trends [23]. In resource-constrained environments, lightweight designs are critical. *MKANet* employs multi-branch kernel-sharing atrous convolutions to achieve large receptive fields without deep stacking, offering an efficient alternative for land-cover classification on edge devices [24]. Additionally, ensemble approaches like *EoS-FM* aggregate multiple lightweight, task-specialized ConvNeXt encoders, achieving balanced generalization superior to single large models while maintaining parameter efficiency [25].

2.2 Self-Supervised Pre-training Paradigms

The core innovation of GFMs lies in their pre-training objectives, which leverage vast amounts of unlabeled satellite data to learn robust representations. The field has evolved from simple reconstruction tasks to sophisticated objectives that exploit the unique multi-view, multi-sensor, and temporal nature of EO data.

2.2.1 Masked Image Modeling (MIM) and Reconstruction Masked Image Modeling, popularized by MAE, remains a dominant pre-training strategy. In the EO domain, standard MIM has been adapted to handle spectral and temporal dimensions. *SpectralGPT* introduces 3D spatial-spectral cube masking, reconstructing missing spectral bands alongside spatial patches to capture inter-band dependencies, permitting aggressive masking ratios (up to 90%) due to high spectral redundancy [26]. *SIGMAE* enhances this by incorporating spectral indices as priors to guide dynamic token masking, prioritizing semantically salient regions during pre-training to improve discriminative power [27].

Temporal extensions of MIM address the dynamic nature of EO. *SatSwinMAE* integrates Video Swin Transformer blocks to process spatiotemporal cubes, reconstructing masked patches across time to learn motion and phenological patterns [16]. *NeighborMAE* proposes joint reconstruction of neighboring image pairs based on geospatial overlap, forcing the model to learn spatial correlations and robust representations despite acquisition variations [28]. Furthermore, *SenPa-MAE* injects sensor parameters (spectral response functions, GSD) directly into patch embeddings, enabling robust zero-shot transfer between sensors with mismatched spectral bands [29].

2.2.2 Contrastive Learning and Instance Discrimination Contrastive learning frameworks, which maximize agreement between augmented views of the same image, have proven effective for learning invariant features. *SatDINO* adapts the DINO framework for remote sensing by enforcing alignment between local and global views with uniform scale sampling, yielding features more robust to multi-scale variations than MAE-based models [30]. *Embedding Earth* extends contrastive learning to the pixel level, enforcing embedding proximity for pixels at identical geographic coordinates across augmented time-series views, effectively leveraging spatiotemporal consistency for dense representation learning [31].

Advanced contrastive strategies exploit the multi-sensor nature of EO data. *X-STARs* combines standard DINO loss with a Multi-Sensor Alignment Dense (MSAD) loss, aligning patch-wise token representations between different sensors (e.g., SAR and Optical) to learn sensor-agnostic features [32]. *SoftCon* utilizes soft contrastive loss guided by multi-label land cover maps, resolving the rigid positive sample definitions of standard InfoNCE loss and enabling smoother latent space distributions [33]. Additionally, *GRAFT* leverages geotagged ground-level imagery as a transitive intermediary to align satellite encoders with CLIP’s text space without direct satellite-text annotations, enabling zero-shot capabilities [34].

2.2.3 Generative and Autoregressive Objectives Beyond reconstruction and contrastive learning, generative and autoregressive objectives are gaining traction. *EarthPT* adapts large language model architectures for EO time series, treating multispectral reflectance values as tokens and training via next-token prediction to enable accurate long-horizon forecasting [35]. *SITS-DECO* employs a minimalist decoder-only transformer, framing all EO tasks as unified generative token sequences, which allows for flexible multi-task learning without architectural modification [36]. In the realm of joint embedding predictive architectures, *SpectralEarth-FM* utilizes a JEPA objective to predict latent representations of global multimodal views from single-sensor local crops, avoiding the noise associated with pixel-level reconstruction across heterogeneous sensors [37].

2.3 Multi-Modal and Cross-Sensor Fusion

A defining characteristic of modern GFMs is their ability to integrate heterogeneous data sources, including optical, SAR, LiDAR, and textual modalities. This fusion enhances robustness to cloud cover, diurnal cycles, and sensor-specific artifacts.

2.3.1 Optical-SAR Fusion and Sensor Agnosticism Fusing Synthetic Aperture Radar (SAR) with optical imagery provides all-weather observation capabilities. *CROMA* and *DOFA* represent state-of-the-art approaches in this domain. *DOFA* (Dynamic One-For-All) employs a wavelength-conditioned dynamic hypernetwork that generates patch embedding weights based on spectral central wavelengths, allowing a single model to dynamically interpolate weights for unseen sensor configurations [9]. *Panopticon* treats multi-sensor snapshots of identical geolocations as natural augmentations, inducing robust spectral and scale invariances through a two-stage spectral progressive pre-training strategy [8].

For scenarios with unpaired data, *CLOSP* employs text as a common bridge to align unpaired SAR and optical imagery in a shared latent space, bypassing the need for pixel-perfect registration [38]. *SkyCap* demonstrates that optical foundation models, when paired with appropriate preprocessing (dB + Z-score), can outperform SAR-specific models on change detection tasks via optical-to-SAR label transfer, highlighting the potential of cross-modal knowledge distillation [39].

2.3.2 Hyperspectral and Multi-Spectral Integration Hyperspectral imagery (HSI) presents unique challenges due to high dimensionality and spectral redundancy. *HyperSIGMA* employs parallel spatial and spectral ViT subnetworks with Sparse Sampling Attention (SSA) to adaptively sample key-value pairs, mitigating redundancy while capturing fine-grained spectral signatures [40]. *SpectralEarth-FM* introduces sensor-specific spectral tokenization, grouping contiguous bands before spatial encoding to preserve spectral continuity [37]. For broader spectral coverage, *Any-Optical-Model* utilizes a spectrum-independent tokenizer with explicit channel index encoding, enabling arbitrary band handling across diverse optical sensors [18].

2.3.3 Vision-Language and Text-Grounded Models Integrating natural language enables interactive querying and zero-shot reasoning. *LISAT* (Language-Instructed Segmentation Assistant) integrates a Vicuna LLM with a Remote-CLIP visual encoder, using an embedding-as-mask paradigm to perform reasoning-driven segmentation based on complex textual queries [41]. *DescribeEarth* employs a scale-adaptive focal strategy and domain-guided fusion to generate object-level captions, balancing local detail with global context [42]. For ultra-high-resolution analysis, *ImageRAG* utilizes a retrieval-augmented generation approach, partitioning images into multi-scale patches and retrieving relevant visual context to answer complex questions without retraining the entire model [43]. Furthermore, *Zero-Shot Multi-Spectral Learning* demonstrates that generalist LMMs can interpret novel spectral modalities when represented as pseudo-images accompanied by explicit textual descriptions of band physics [44, 45].

2.4 Temporal Modeling and Time-Series Analysis

Satellite image time series (SITS) are critical for monitoring phenology, urban growth, and disaster response. GFMs must effectively model irregular temporal sampling and long-range dependencies.

2.4.1 Regular and Irregular Time-Series Encoding Standard transformers often assume regular time intervals, which is rarely the case in EO due to cloud cover and orbital constraints. *ALISE* (ALigned Sits Encoder) introduces a learnable query-based temporal projector that maps irregular input sequences onto fixed-size aligned latent spaces, enabling direct linear probing without fine-tuning [46]. *TESSERA* processes pixel-level annual time series (‘d-pixels’) via parallel Transformer encoders with Day-of-Year positional encoding, learning representations invariant to cloud cover through random temporal sampling [47].

2.4.2 Phenology-Aware and Seasonal Sampling Effective pre-training requires diverse temporal samples. *SSL4Eco* replaces fixed calendar windows with phenology-informed seasonal sampling based on Enhanced Vegetation Index (EVI) stages, capturing diverse vegetation states more effectively than random sampling [48]. *HighFM* adapts the SatMAE framework with fine-grained temporal encodings retaining minute-level timestamp data, crucial for tracking rapid events like wildfires using geostationary data [49]. *ViTs for SITS* factorizes processing into sequential temporal and spatial encoders, optimizing the order to extract robust growth cycle features before aggregating spatial evidence [50].

2.5 Parameter-Efficient Fine-Tuning and Adaptation

Given the massive size of GFMs, full fine-tuning is often computationally prohibitive. Parameter-Efficient Fine-Tuning (PEFT) methods have become essential for adapting these models to downstream tasks.

2.5.1 Low-Rank Adaptation (LoRA) and Adapters LoRA has been successfully adapted for dense prediction tasks in EO. *LoRA-adapted Clay* integrates low-rank matrices into the query-key-value layers of the Clay Vision Transformer, achieving higher segmentation accuracy than frozen encoders while avoiding out-of-memory errors [51]. *PEFT for RS-VG* systematically evaluates LoRA placement in visual grounding models, finding that adapting both image encoder and decoder yields superior performance with minimal parameter updates (3.26%) [52]. *DEFLECT* proposes an Embedding Deflection mechanism that disentangles spectral and geometric embeddings, allowing

frozen pretrained parameters to handle spatial structures while minimal trainable parameters adapt to spectral variations [53].

2.5.2 Knowledge Distillation and Model Compression Distillation is critical for deploying GFMs on resource-constrained platforms. *InstaGeo* employs task-specific distillation where student models retain only the necessary encoder depth, achieving up to 8x parameter reduction with negligible performance loss [54]. *DualKD* utilizes a dual-teacher framework with dynamic weighting based on teacher confidence to guide student models, mitigating noise from unreliable teachers [55]. For onboard deployment, *First On-Orbit Demonstration* compresses a 300M parameter ViT teacher into a 256-MAE-D student via dual-MAE distillation, achieving a 16x size reduction suitable for spaceborne inference [13]. *How Much of a Model Do We Need?* reveals that RS FMs retain >71% relative accuracy at 1% compute budget, suggesting significant inherent redundancy that can be exploited for slimming [14].

2.6 Generative World Models and Image Synthesis

Generative models are increasingly used not just for data augmentation but as foundational components for simulation and forecasting.

2.6.1 Diffusion Models for EO Diffusion models offer superior stability and control compared to GANs. *DiffusionSat* extends Stable Diffusion with 3D ControlNets and sinusoidal embeddings of numerical metadata (latitude, longitude, timestamp), enabling precise geospatial and temporal conditioning for image generation [56]. *MetaEarth* implements a resolution-guided self-cascading framework, recursively generating high-resolution images from low-resolution predecessors to preserve global semantics while refining local details [57]. *Sat-JEPA-Diff* decouples forecasting into semantic prediction and texture synthesis, using an IJEPA module to predict future latent embeddings which then condition a frozen diffusion backbone [58].

2.6.2 World Models and Forecasting The concept of “World Models” is emerging in EO, aiming to simulate future states of the Earth system. *RS-WorldModel* unifies spatiotemporal change understanding and future scene forecasting via an autoregressive vision-language framework, trained with geo-aware generative pre-training [59]. *RemoteBAGEL* formulates remote sensing world modeling as direction-conditioned spatial extrapolation, effectively modeling geospatial transitions when trained on overlapping grid trajectories [60]. These models represent a shift from static perception to dynamic, predictive reasoning about Earth processes.

3. Challenges and Outlook

Despite rapid progress, the field of Geospatial Foundation Models faces significant hurdles that must be addressed to realize their full potential.

3.1 Robustness and Out-of-Distribution Generalization

A critical finding in recent literature is the lack of robustness in current GFMs. *EarthShift* demonstrates that geospatial foundation models exhibit no robustness advantage over generic ImageNet-pretrained models when faced with realistic natural distribution shifts (e.g., geographic, temporal, sensor shifts) [11]. Similarly, *Rethinking Electro-Optical Vision Foundation Models* shows that

EO-specific VFMs suffer significant performance degradation under cross-scene settings, often underperforming generalist models like DINOv3 [12]. *Do Foundation Model Embeddings Improve Cross-Country Crop Yield Generalisation?* further highlights that frozen embeddings offer no advantage over simple spectral medians in leave-one-country-out evaluations due to severe domain shifts [61]. Future research must prioritize objectives that explicitly enforce invariance to these shifts, potentially through causal inference frameworks [62, 63] and diverse, globally representative pre-training data [64].

3.2 Physical Consistency and Scientific Plausibility

Current GFMs often neglect explicit physical constraints, leading to scientifically implausible predictions. *From Physics to Foundation Models* notes that while FMs offer unified representation learning, they currently rely on shallow regression heads lacking task-specific structure, resulting in violations of physical laws (e.g., mass conservation, radiative transfer) [65]. *On the Foundations of Earth and Climate Foundation Models* proposes that ideal models must incorporate physical consistency as a core feature, potentially through physics-informed loss functions or hybrid neural-symbolic architectures [66]. Integrating domain knowledge, such as process convolutions and latent force models, remains an underexplored avenue for enhancing the scientific validity of GFMs [67].

3.3 Evaluation Bottlenecks and Benchmarking

The field suffers from fragmented evaluation protocols. While benchmarks like *GEO-Bench*, *PhilEO Bench*, and *Landsat-Bench* provide standardized frameworks, they often exclude critical dimensions such as temporal fine-tuning, multi-modal fusion, and extreme event detection [68, 69, 70]. *ExEBench* attempts to address this by integrating EO and weather data for extreme event tasks, revealing that models often under-predict extreme temperatures and suffer from geographic bias [71]. Furthermore, *Revisiting pre-trained remote sensing model benchmarks* emphasizes that preprocessing choices (resizing, normalization) significantly impact reported performance, necessitating strict standardization to ensure fair comparisons [72]. Future benchmarks must encompass a broader range of tasks, including causal inference, long-horizon forecasting, and interactive reasoning [73, 74].

3.4 Deployment and Operational Constraints

Deploying GFMs in operational settings, particularly onboard satellites, presents unique challenges. *Satellite edge computing* studies highlight the trade-offs between communication and processing costs, suggesting that distributing segments across neighboring satellites via Inter-Satellite Links can alleviate bottlenecks [75]. However, thermal throttling, memory constraints, and the need for real-time inference limit the complexity of models that can be deployed [13, 76]. *Deploying Geospatial Foundation Models in the Real World* outlines a structured protocol for operational deployment, emphasizing the importance of geographic and temporal holdout testing to ensure reliability in production environments [77]. Advances in quantization, pruning, and neuromorphic computing (e.g., Spike Neural Networks) will be crucial for enabling efficient edge intelligence [78, 79].

3.5 Future Directions: Agentic AI and Multimodal Reasoning

The future of EO lies in the transition from passive perception to active, agentic reasoning. *Earth AI* and *RemoteReasoner* demonstrate the potential of multi-agent orchestration and reinforcement

learning to decompose complex geospatial queries and execute multi-step reasoning workflows [80, 81]. Integrating GFMs with large language models and tool-use capabilities will enable autonomous systems for disaster response, resource management, and scientific discovery [74]. Additionally, the integration of raster and vector data into unified representation spaces (*Spatial Representation Learning Beyond Pixels*) promises to bridge the gap between pixel-level analysis and human-centric geospatial semantics [82].

4. Conclusion

Foundation Models have fundamentally altered the landscape of Satellite Earth Observation, offering unprecedented capabilities for learning from vast, unlabeled data archives. Through architectural innovations like State Space Models and hierarchical transformers, and pre-training strategies that exploit the multi-sensor, multi-temporal nature of EO data, these models are achieving state-of-the-art performance across a wide array of tasks. However, significant challenges remain in ensuring robustness to distribution shifts, maintaining physical consistency, and enabling efficient deployment in resource-constrained environments. As the field matures, the focus must shift from merely scaling model size to developing smarter, more efficient, and scientifically grounded architectures. The integration of agentic reasoning, causal inference, and unified raster-vector representations will define the next generation of Geospatial AI, ultimately empowering humanity to better understand and manage our planet.

References

- [1] Maxim Neumann, Andre Susano Pinto, Xiaohua Zhai. (2019). TRAINING GENERAL REPRESENTATIONS FOR REMOTE SENSING USING IN-DOMAIN KNOWLEDGE. arXiv:2010.00332.
- [2] Saad Lahrichi, Zion Sheng, Shufan Xia. (2024). Is Self-Supervised Pre-training on Satellite Imagery Better than ImageNet? A Systematic Study with Sentinel-2. arXiv:2502.10669.
- [3] Nikolaos Dionelis, Jente Bosmans, Riccardo Musto. (2024). Scaling-Up the Pretraining of the Earth Observation Foundation Model PhilEO to the MajorTOM Dataset. arXiv:2506.14765.
- [4] Charith Wickrema, Eliza Mace, Hunter Brown. (2025). Scaling Remote Sensing Foundation Models: Data Domain Tradeoffs at the Peta-Scale. arXiv:2512.23903.
- [5] Yi Wang, Nassim Ait Ali Braham, Zhitong Xiong. (2022). SSL4EO-S12: A Large-Scale Multi-Modal, Multi-Temporal Dataset for Self-Supervised Learning in Earth Observation. arXiv:2211.07044.
- [6] Yi Wang, Conrad M Albrecht, Nassim Ait Ali Braham. (2022). Self-supervised Learning in Remote Sensing: A Review. arXiv:2206.13188.
- [7] Ani Vanyan, Alvard Barseghyan, Hakob Tamazyan. (2024). Do Satellite Tasks Need Special Pretraining?. arXiv:2510.17014.
- [8] Leonard Waldmann, Ando Shah, Yi Wang. (2024). Panopticon: Advancing Any-Sensor Foundation Models for Earth Observation. arXiv:2503.10845.
- [9] Zhitong Xiong, Yi Wang, Fahong Zhang. (2024). Neural Plasticity-Inspired Multimodal Foundation Model for Earth Observation. arXiv:2403.15356.

- [10] Kevin Lane, Morteza Karimzadeh. (2025). A Genealogy of Multi-Sensor Foundation Models in Remote Sensing. arXiv:2504.17177.
- [11] Kelsey Doerksen, Hannah Kerner. (2024). EarthShift: a benchmark for measuring robustness to real-world distribution shifts in Earth observation. arXiv:2605.29330.
- [12] Hyobin Park, Minseok Seo, Dong-Geol Choi. (2024). Rethinking Electro-Optical Vision Foundation Models for Remote Sensing Retrieval: A Controlled Comparison with Generalist VFMs. arXiv:2605.02283.
- [13] Andrew Du, Roberto Del Prete, Alejandro Mousist. (2025). First On-Orbit Demonstration of a Geospatial Foundation Model. arXiv:2512.01181.
- [14] Leonard Hackel, Tom Burgert, Begüm Demir. (2025). How Much of a Model Do We Need? Redundancy and Slimmability in Remote Sensing Foundation Models. arXiv:2601.22841.
- [15] Keumgang Cha, Junghoon Seo, Taekyung Lee. (2024). A Billion-scale Foundation Model for Remote Sensing Images. arXiv:2304.05215.
- [16] Yohei Nakayama, Jiawei Su, Luis M. Pazos-Outón. (2024). SatSwinMAE: Efficient Autoencoding for Multiscale Time-series Satellite Imagery. arXiv:2405.02512.
- [17] Eric Guérin, Killian Oechsli, Christian Wolf. (2021). Satellite Image Semantic Segmentation. arXiv:2110.05812.
- [18] Xuyang Li, Chenyu Li, Danfeng Hong. (2025). Any-Optical-Model: A Universal Foundation Model for Optical Remote Sensing. arXiv:2512.17224.
- [19] Patrick Kage, Pavlos Andreadis. (2024). MULTI-MODAL, MULTI-SCALE REPRESENTATION LEARNING FOR SATELLITE IMAGERY ANALYSIS JUST NEEDS A GOOD ALIBI. arXiv:2604.10347.
- [20] Chuc Man Duc, Hiromichi Fukui. (2024). SatMamba: Development of Foundation Models for Remote Sensing Imagery Using State Space Models. arXiv:2502.00435.
- [21] Fengxiang Wang, Hongzhen Wang, Yulin Wang. (2024). RoMA: Scaling up Mamba-based Foundation Models for Remote Sensing. arXiv:2503.10392.
- [22] Jules Decaestecker, Nicolas Vigne. (2024). PathMamba: A Hybrid Mamba-Transformer for Topologically Coherent Road Segmentation in Satellite Imagery. arXiv:2511.21298.
- [23] Paul Vermeeren, Heysem Kaya. (2024). Developing a foundation model for high-resolution remote sensing data of the Netherlands. arXiv:2605.10184.
- [24] Zhiqi Zhang, Wen Lu, Jinshan Cao. (2024). MKANet: A Lightweight Network with Sobel Boundary Loss for Efficient Land-cover Classification of Satellite Remote Sensing Imagery. arXiv:2207.13866.
- [25] Pierre Adorni, Minh-Tan Pham, Stephane May. (2025). EoS-FM: Can an Ensemble of Specialist Models act as a Generalist Feature Extractor?. arXiv:2511.21523.
- [26] Danfeng Hong, Bing Zhang, Xuyang Li. (2024). SpectralGPT: Spectral Remote Sensing Foundation Model. arXiv:2311.07113.
- [27] Xiaokang Zhang, Bo Li, Chufeng Zhou. (2025). SIGMAE: A Spectral-Index-Guided Foundation Model for Multispectral Remote Sensing. arXiv:2603.07463.

- [28] Liang Zeng, Valerio Marsocci, Wufan Zhao. (2024). NeighborMAE: Exploiting Spatial Dependencies between Neighboring Earth Observation Images in Masked Autoencoders Pretraining. arXiv:2603.02522.
- [29] Jonathan Prexl, Michael Schmitt. (2024). SenPa-MAE: Sensor Parameter Aware Masked Autoencoder for Multi-Satellite Self-Supervised Pretraining. arXiv:2408.11000.
- [30] Jakub Straka, Ivan Gruber. (2024). SatDINO: A Deep Dive into Self-Supervised Pretraining for Remote Sensing. arXiv:2508.21402.
- [31] Michail Tarasiou, Stefanos Zafeiriou. (2021). Embedding Earth: Self-supervised contrastive pre-training for dense land cover classification. arXiv:2203.06041.
- [32] Valerio Marsocci, Nicolas Audebert. (2024). Cross-sensor self-supervised training and alignment for remote sensing. arXiv:2405.09922.
- [33] Yi Wang, Conrad M Albrecht, Xiao Xiang Zhu. (2024). Multi-label Guided Soft Contrastive Learning for Efficient Earth Observation Pretraining. arXiv:2405.20462.
- [34] Utkarsh Mall, Cheng Perng Phoo, Meilin Kelsey Liu. (2024). Remote Sensing Vision-Language Foundation Models Without Annotations via Ground Remote Alignment. arXiv:2312.06960.
- [35] Michael J. Smith, Luke Fleming, James E. Geach. (2024). EarthPT: a time series foundation model for Earth Observation. arXiv:2309.07207.
- [36] Samuel J. Barrett, Docko Sow. (2025). SITS-DECO: A GENERATIVE DECODER IS ALL YOU NEED FOR MULTITASK SATELLITE IMAGE TIME SERIES MODELLING. arXiv:2510.21813.
- [37] Nassim Ait Ali Braham, Aaron Banze, Conrad M. Albrecht. (2024). SpectralEarth-FM: Bringing Hyperspectral Imagery into Multimodal Earth Observation Pretraining. arXiv:2605.21075.
- [38] Daniele Rege Cambrin, Lorenzo Vaiani, Giuseppe Gallipoli. (2024). Text-to-Remote-Sensing-Image Retrieval beyond RGB Sources. arXiv:2507.10403.
- [39] Paul Weinmann, Ferdinand Schenck, Martin Siklar. (2025). SkyCap: Bitemporal VHR Optical-SAR Quartets for Amplitude Change Detection and Foundation-Model Evaluation. arXiv:2512.14755.
- [40] Di Wang, Meiqi Hu, Yao Jin. (2024). HyperSIGMA: Hyperspectral Intelligence Comprehension Foundation Model. arXiv:2406.11519.
- [41] Jerome Quenum, Wen-Han Hsieh, Tsung-Han Wu. (2024). LISAT: Language-Instructed Segmentation Assistant for Satellite Imagery. arXiv:2505.02829.
- [42] Kaiyu Li, Zixuan Jiang, Xiangyong Cao. (2024). DescribeEarth: Describe Anything for Remote Sensing Images. arXiv:2509.25654.
- [43] Zilun Zhang, Haozhan Shen, Tiancheng Zhao. (2024). Enhancing Ultra High Resolution Remote Sensing Imagery Analysis with ImageRAG. arXiv:2411.07688.
- [44] Ganesh Mallya, Yotam Gigi, Dahun Kim. (2024). Zero-Shot Multi-Spectral Learning: Reimagining a Generalist Multimodal Gemini 2.5 Model for Remote Sensing Applications. arXiv:2509.19087.

- [45] Dahun Kim, Ganesh Satish Mallya, Anelia Angelova. (2024). Unlocking Multi-Spectral Data for Multi-Modal Models with Guided Inputs and Chain-of-Thought Reasoning. arXiv:2604.21032.
- [46] Iris Dumeur, Silvia Valero, Jordi Inglada. (2024). Paving the way toward foundation models for irregular and unaligned Satellite Image Time Series. arXiv:2407.08448.
- [47] Zhengpeng Feng, Clement Atzberger, Sadiq Jaffer. (2025). TESSERA: Temporal Embeddings of Surface Spectra for Earth Representation and Analysis. arXiv:2506.20380.
- [48] Damien Robert, Johannes Dollinger, Elena Plekhanova. (2024). SSL4Eco: A Global Seasonal Dataset for Geospatial Foundation Models in Ecology. arXiv:2504.18256.
- [49] Stella Girtsou, Konstantinos Alexis, Giorgos Giannopoulos. (2025). HighFM: Towards a Foundation Model for Learning Representations from High-Frequency Earth Observation Data. arXiv:2604.04306.
- [50] Michail Tarasiou, Erik Chavez, Stefanos Zafeiriou. (2021). ViTs for SITS: Vision Transformers for Satellite Image Time Series. arXiv:2301.04944.
- [51] Karthick Panner Selvam, Raúl Ramos-Pollán, Freddie Kalaitzis. (2024). Rapid Adaptation of Earth Observation Foundation Models for Segmentation. arXiv:2409.09907.
- [52] Hasan Moughnieh, Mohamad Chalhoub, Hasan Nasrallah. (2024). EFFICIENT ADAPTATION FOR REMOTE SENSING VISUAL GROUNDING. arXiv:2503.23083.
- [53] Romain Thoreau, Valerio Marsocci, Dawa Derksen. (2024). Parameter-Efficient Adaptation of Geospatial Foundation Models through Embedding Deflection. arXiv:2503.09493.
- [54] Ibrahim Salihu Yusuf, Iffanice Houndayi, Rym Oualha. (2024). InstaGeo: Compute-Efficient Geospatial Machine Learning from Data to Deployment. arXiv:2510.05617.
- [55] Thanh-Dung Le, Vu Nguyen Ha, Ti Ti Nguyen. (2024). Semantic Knowledge Distillation for Onboard Satellite Earth Observation Image Classification. arXiv:2411.00209.
- [56] Samar Khanna, Patrick Liu, Linqi Zhou. (2024). DiffusionSat: A Generative Foundation Model for Satellite Imagery. arXiv:2312.03606.
- [57] Zhiping Yu, Chenyang Liu, Liqin Liu. (2024). MetaEarth: A Generative Foundation Model for Global-scale Remote Sensing Image Generation. arXiv:2405.13570.
- [58] Kursat Komurcu, Linas Petkevicius. (2025). SAT-JEPA-DIFF: BRIDGING SELF-SUPERVISED LEARNING AND GENERATIVE DIFFUSION FOR REMOTE SENSING. arXiv:2603.13943.
- [59] Linrui Xu, Zhongan Wang, Fei Shen. (2024). RS-WorldModel: a Unified Model for Remote Sensing Understanding and Future Sense Forecasting. arXiv:2603.14941.
- [60] Yuxi Lu, Biao Wu, Zhidong Li. (2025). REMOTE SENSING-ORIENTED WORLD MODEL. arXiv:2509.17808.
- [61] Yaw Osei Adjei. (2024). Do Foundation Model Embeddings Improve Cross-Country Crop Yield Generalisation? A Leave-One-Country-Out Evaluation in Sub-Saharan Africa. arXiv:2605.08113.
- [62] Praveen Ravirathinam, Ankush Khandelwal, Rahul Ghosh. (2024). A Causally Informed Pretraining Approach for Multimodal Foundation Models: Applications in Remote Sensing. arXiv:2407.19660.

- [63] Kazuki Sakamoto, Connor T. Jerzak, Adel Daoud. (2025). A Scoping Review of Earth Observation and Machine Learning for Causal Inference: Implications for the Geography of Poverty. arXiv:2406.02584.
- [64] Mirali Purohit, Gedeon Muhawenayo, Esther Rolf. (2024). How Does the Spatial Distribution of Pre-training Data Affect Geospatial Foundation Models?. arXiv:2501.12535.
- [65] Zhenyu Yu, Mohd Yamani Idna Idris, Hua Wang. (2024). From Physics to Foundation Models: A Review of AI-Driven Quantitative Remote Sensing Inversion. arXiv:2507.09081.
- [66] Xiao Xiang Zhu, Zhitong Xiong, Yi Wang. (2024). On the Foundations of Earth and Climate Foundation Models. arXiv:2405.04285.
- [67] Daniel Heestermans Svendsen, Maria Piles, Jordi Munoz-Mari. (2020). Integrating Domain Knowledge in Data-driven Earth Observation with Process Convolutions. arXiv:2104.08134.
- [68] Alexandre Lacoste, Hannah Kerner, Hamed Alemohammad. (2023). GEO-Bench: Toward Foundation Models for Earth Monitoring. arXiv:2306.03831.
- [69] Isaac Corley, Lakshay Sharma, Ruth Crasto. (2025). Landsat-Bench: Datasets and Benchmarks for Landsat Foundation Models. arXiv:2506.08780.
- [70] Casper Fibaek, Luke Camilleri, Andreas Luyts. (2024). PhilEO BENCH: EVALUATING GEO-SPATIAL FOUNDATION MODELS. arXiv:2401.04464.
- [71] Shan Zhao, Zhitong Xiong, Jie Zhao. (2024). ExEBench: Benchmarking Foundation Models on Extreme Earth Events. arXiv:2505.08529.
- [72] Isaac Corley, Caleb Robinson, Rahul Dodhia. (2023). Revisiting pre-trained remote sensing model benchmarks: resizing and normalization matters. arXiv:2305.13456.
- [73] Mojtaba Valipour, Kelly Zheng, James Lowman. (2024). AGI for the Earth, the path, possibilities and how to evaluate intelligence of models that work with Earth Observation Data?. arXiv:2508.06057.
- [74] Xiangyu Zhao, Bo Liu, Yuehan Zhang. (2024). Earth Science Foundation Models: From Perception to Reasoning and Discovery. arXiv:2605.12542.
- [75] Israel Leyva-Mayorga, Marc M. Gost, Marco Moretti. (2024). Satellite edge computing for real-time and very-high resolution Earth observation. arXiv:2212.12912.
- [76] Beatriz Soret, Antonio M. Mercado-Martínez, Antonio Jurado-Navas. (2026). Edge Intelligence for Satellite-based Earth Observation: Scheduling Image Acquisition and Processing. arXiv:2604.05937.
- [77] Christina Butsko, Kristof Van Tricht, Gabriel Tseng. (2025). Deploying Geospatial Foundation Models in the Real World: Lessons from WorldCereal. arXiv:2508.00858.
- [78] Chaabane Djeraba, Jérôme Riedi. (2024). Mining atmospheric data. arXiv:2106.13992.
- [79] Beatriz Soret, Israel Leyva-Mayorga, Antonio M. Mercado-Martínez. (2024). Semantic and goal-oriented edge computing for satellite Earth Observation. arXiv:2408.15639.
- [80] Aaron Bell, Amit Aides, Amr Helmy. (2025). Earth AI: Unlocking Geospatial Insights with Foundation Models and Cross-Modal Reasoning. arXiv:2510.18318.

- [81] Liang Yao, Fan Liu, Hongbo Lu. (2025). RemoteReasoner: Towards Unifying Geospatial Reasoning Workflow. arXiv:2507.19280.
- [82] Steffen Knoblauch, Hao Li, Gengchen Mai. (2024). Spatial Representation Learning Beyond Pixels: Unifying Raster Data and Vector Semantics for Human-Centric Geospatial Foundation Models. arXiv:2606.02374.